HW2 (Kernel Density)

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2024-06-20

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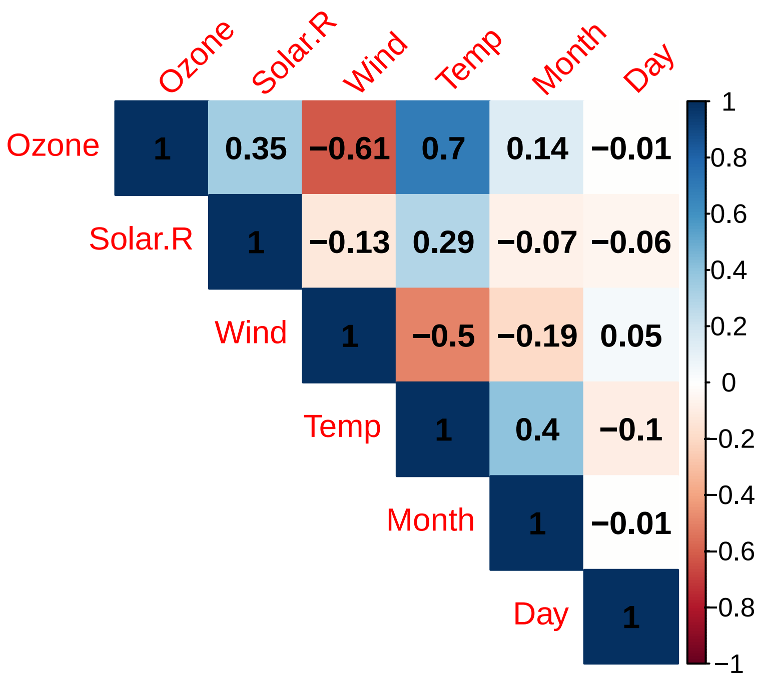
```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)
```

Question 1

1. Find the correlation matrix for Ozone, Solar R, Wind, and Temperature by excluding missing values.

```
data(airquality)
airquality_cc <- na.omit(airquality)
# library(Hmisc)
# latex(describe(airquality_cc), file = "", caption.placement = "top")
```

```
library(corrplot)
M <- cor(airquality_cc)
corrplot(M,  method = 'color', addCoef.col = 'black', type = 'upper',
         tl.srt = 45, diag = T)
```



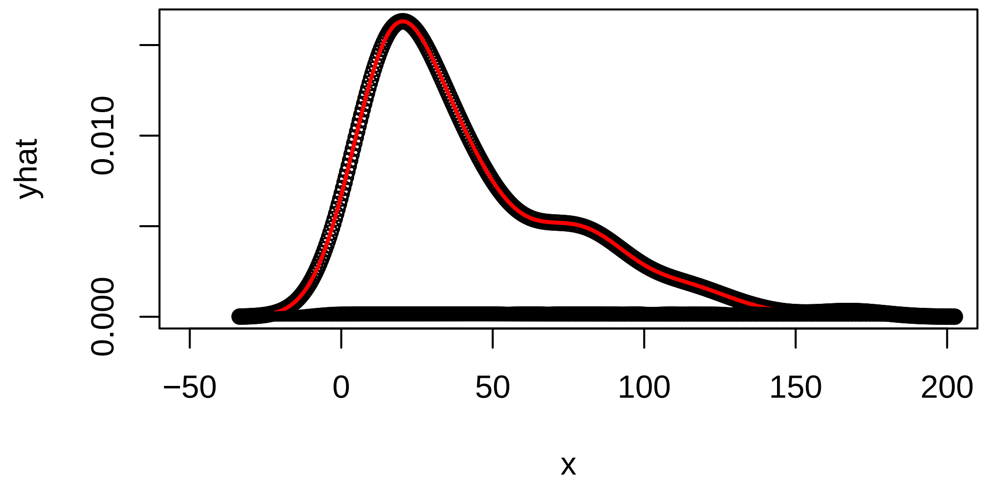
From the correlation matrix plot, we can find that `Temp` and `Ozone` has strongly positive linear relationship with correlation coefficient $\rho = 0.7$. On the other hand, `Wind` has negative relationship with `Ozone` and `Temp`, with $\rho = -0.61$ and -0.5.

2. Extract "Ozone" data vector. Plot the histogram with a step-by-step kernel density estimate.

```
x <- na.omit(airquality_cc$Ozone)
myt <- density(x, kernel = "gaussian")$x
```

```
n <- length(x)
bw <- 11.52194
ft <- matrix(NA, ncol = length(myt), nrow = n)
for (i in 1:n){
  ft[i, ] <- dnorm( (myt-x[i])/bw )/(n*bw)
}
yhat <- colSums(ft)
```

```
plot(x = myt, y = yhat, xlim = c(-50,200), cex=1, type = "p", col="black", xlab="x")
lines(density(x, kernel = "gaussian"), col="red", lwd=2)
for (i in 1:length(x)) points(x = myt, y = ft[i,], xlim = c(-50,200), cex=0.5)
```



3. Based on question (2), use gamma distribution to fit the dataset "Ozone" and plot it. What's the estimations of the shape and scale parameters? Does it close to the kernel density plot?

Answer:

```
library(MASS)
# estimated shape and scale
gamma.par <- fitdistr(x, dgamma, list(shape = 1, scale = 25))
print(gamma.par)
```

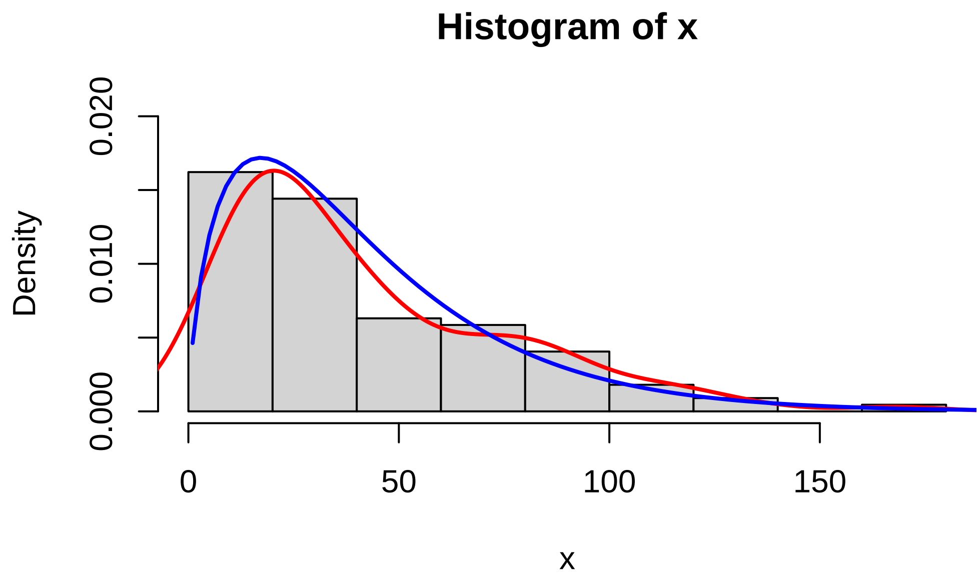
```
     shape         scale
   1.6893397   24.9271249
 ( 0.2083877) ( 3.5749683)
```

The estimations of shape parameter and scale parameters are $\hat{\alpha} = 1.689$ with $sd = 0.208$ and $\hat{\beta} = 24.927$ with $sd = 3.575$.

```
est.shape = gamma.par$estimate[1]
est.scale = gamma.par$estimate[2]

hist(x, probability = T, ylim = c(0,0.02))
```

```r
lines(density(x, kernel = "gaussian"), col = "red", lwd = 2)
curve(dgamma(x, shape = est.shape, scale = est.scale),from = 1, to = 200, add = T, lwd = 2, c
```

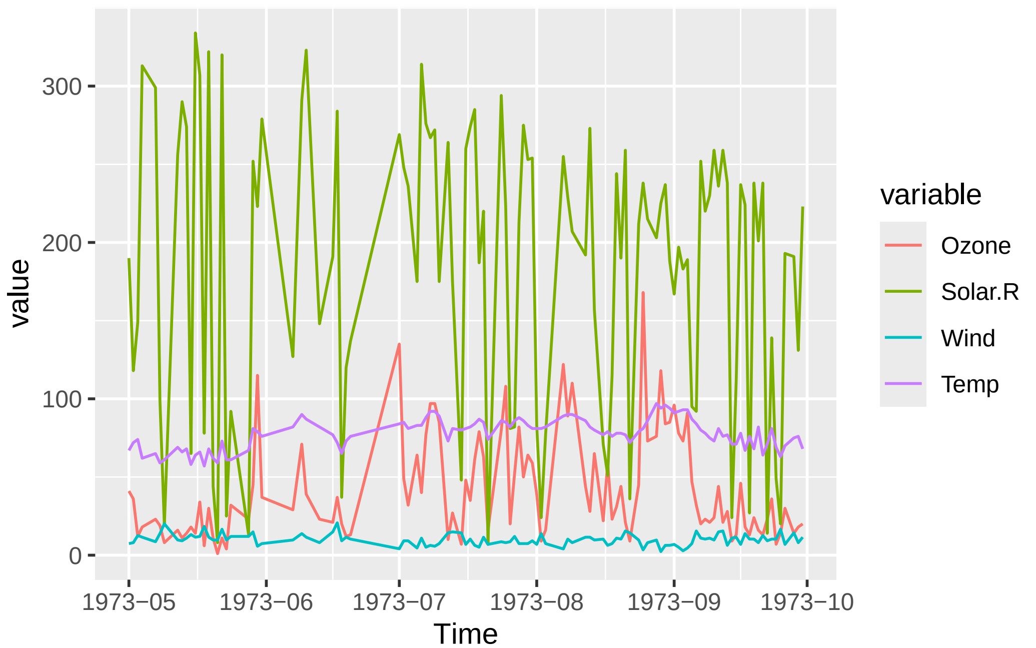


From the plot abova, the estimated gamma distribution is closed to kernel density distribution.

4. Please show time series plots based on daily readings of Ozone, Solar R, Wind, and Temperature for May 1 to September 30. What is your finding?

```r
library(TSA)
library(ggplot2)
library(xts)
library(dygraphs)
library(reshape2)
library(lubridate)
dat <- paste(as.character(airquality_cc$Month),"/",
             as.character(airquality_cc$Day), sep="")
dat <- as.Date(dat, format = "%m/%d") # default year = 2024.
year(dat) <- 1973 # Change year to 1973
df <- data.frame(Ozone = airquality_cc$Ozone,
                 Solar.R = airquality_cc$Solar.R,
                 Wind = airquality_cc$Wind,
                 Temp = airquality_cc$Temp,
```

```r
                  Time = ymd(dat))
data <- melt(df, id.vars = "Time")
ggplot(data, aes(x = Time, y = value, col = variable)) + geom_line()+
  scale_x_date(date_labels = "%Y-%m")
```



Question 2

Let the kernel function $K(x)$ is a probability density symmetric around zero. Show that a kernel density estimator

$$\hat{f}_n(x) = \sum_{i=1}^{n} \frac{K(\frac{x-x_i}{h_n})}{nh_n}$$

is a probability density function. pf :

$$\int_{-\infty}^{\infty} \hat{f}_n(x)dx = \int_{-\infty}^{\infty} \sum_{i=1}^{n} \frac{K(\frac{x-x_i}{h_n})}{nh_n} dx$$

$$= \sum_{i=1}^{n} \int_{-\infty}^{\infty} \frac{K(\frac{x-x_i}{h_n})}{nh_n} dx = \sum_{i=1}^{n} \frac{1}{n} \int_{-\infty}^{\infty} \frac{K(\frac{x-x_i}{h_n})}{h_n} dx = \sum_{i=1}^{n} \frac{1}{n} \times 1 = 1.$$